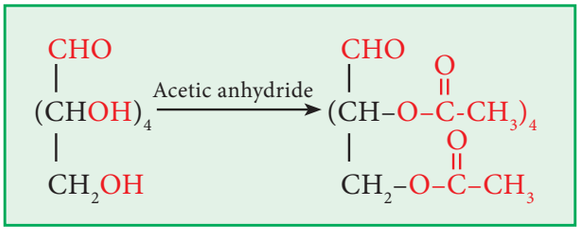
Both glucose and fructose behave as weak acids and form salts with $Ca(OH)_2$ (lime water).

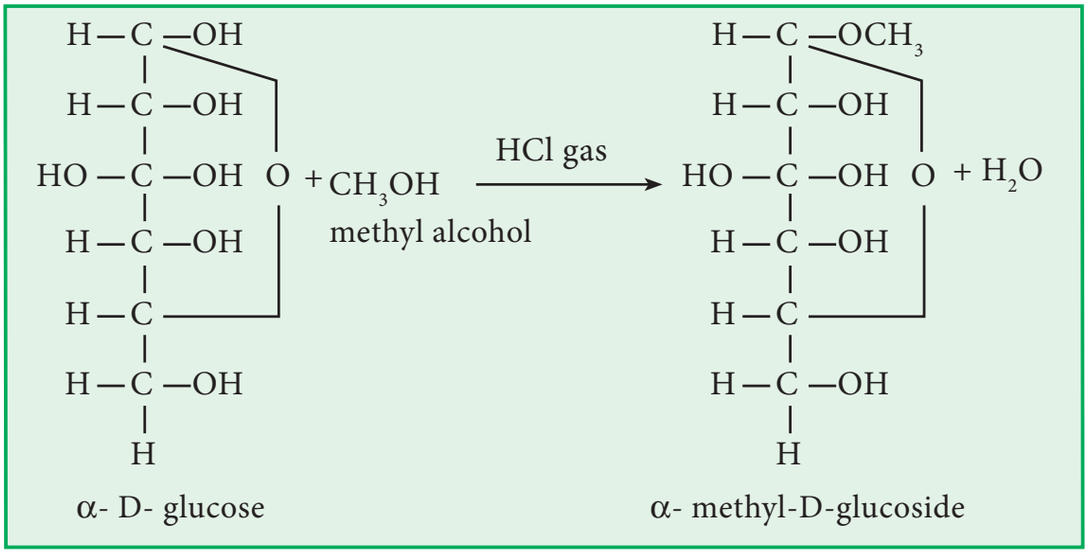
ii. Ester formation:

Glucose and Fructose form penta acetyl derivative when treated with acetic anhydride.



iii. Ether formation:

Glucose and fructose react with methanol in the presence of dry HCl gas to give ethers known as methyl glucoside and methyl fructoside, respectively.



iv. Reduction:

i. Sodium amalgum reduces glucose into sorbitol and fructose into a mixture of sorbitol and mannitol. Similarly, it reduces fructose into a mixture of sorbitol and mannitol.

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